2.4 Boolean Logic

Keyword	Definition
AND	Has two inputs and outputs 1 only if both are 1.
OR	Has two inputs, if both are 1 then outputs 1, otherwise outputs 0.
NOT	Has a single input and outputs the opposite of that input.
Truth Tables	A table which shows the possible input and output values for a logic gate.
Logic Gate	An operator such as AND, OR or NOT which performs a logical function on the input.
Logic Diagram	A diagram showing one or more logic gates together.

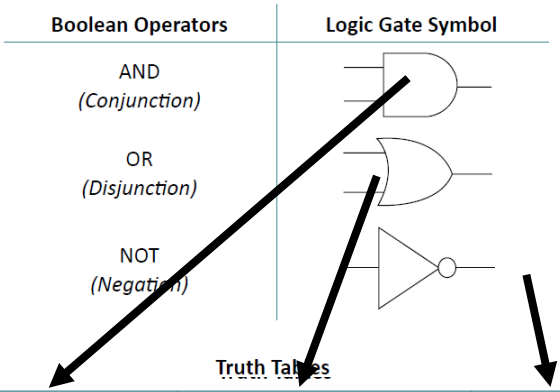
2.4.1 Boolean Logic

2.4.1 Boolean logic

- ☐ Simple logic diagrams using the operators AND, OR and NOT
- ☐ Truth tables
- ☐ Combining Boolean operators using AND, OR and NOT
- ☐ Applying logical operators in truth tables to solve problems

Required

- ✓ Knowledge of the truth tables for each logic gate
- ✓ Recognition of each gate symbol
- ✓ Understanding of how to create, complete or edit logic diagrams and truth tables for given scenarios
- ✓ Ability to work with more than one gate in a logic diagram



Truth Tables

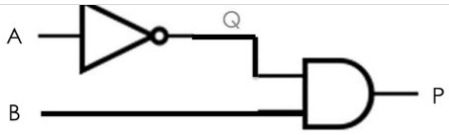
AND			OR			NOT	
A	B	A AND B	A	B	A OR B	A	NOT A
0	0	0	0	0	0	0	1
0	1	0	0	1	1	1	0
1	0	0	1	0	1		
1	1	1	1	1	1		

Alternatives

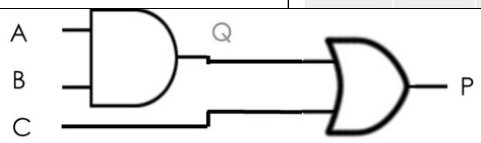
- Use of other valid notation will be accepted within the examination, e.g. Using T/F for 1/0, or V for OR, etc.



A	B	Q = A AND B	P = NOT (A AND B)
1	1	1	0
1	0	0	1
0	1	0	1
0	0	0	1

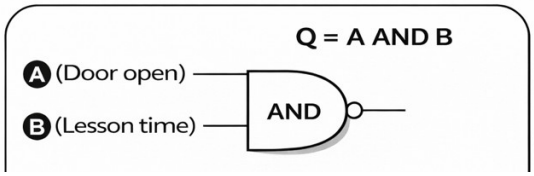


A	B	Q = Not A	P = (NOT A) AND B
1	1	0	0
1	0	0	0
0	1	1	1
0	0	1	0

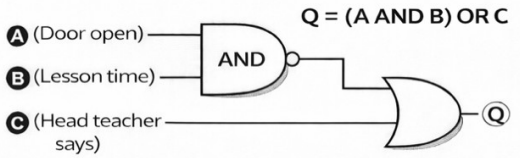


A	B	C	Q = A AND B	P = (A AND B) OR C
1	1	1	1	1
1	1	0	1	1
1	0	1	0	1
1	0	0	0	0
0	1	1	0	1
0	1	0	0	0
0	0	1	0	1
0	0	0	0	0

The alarm sounds only if the door is opened AND it is during lesson time



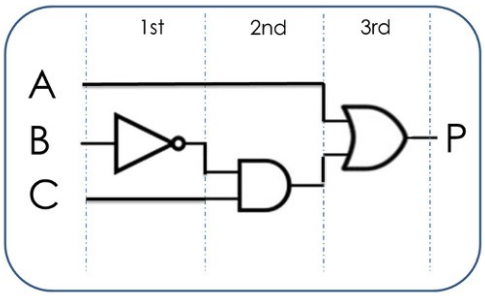
The alarm sounds if the door is opened AND it is during lesson time OR the head teacher says



- **A** = Door is open
- **B** = Break time
- **C** = Headteacher override switch is pressed

The alarm (P) should sound **if the door is open AND it is NOT break time**, **OR** if the headteacher override switch is pressed

So P = ((NOT B) AND C) OR A



2.4.1 Exam Questions on Boolean Logic

Question	Tip to answer
1(a) Name one Boolean operator. **(1)** AND / OR / NOT.	Just name one.
1(b) What output does a NOT gate produce? **(1)** **0 -> 1 1->0**	"Flips it."
1(c) How many rows are in a truth table with 3 inputs? **(1)** 8	Use (2^n).
2(a) Complete the truth table for **P = A AND B**. **(2)**	Only 1 when both 1.

A B → P:
0 0 → 0.¹
0 1 → 0.²
1 0 → 0.¹
1 1 → 1.²

2(b) Complete the truth table for **Q = A OR B**. **(2)**	1 if either is 1.

0 0 → 0.¹
0 1 → 1.²
1 0 → 1.¹
1 1 → 1.²

2(c) Complete the truth table for **R = NOT A**. **(2)**

A=0 → R=1.¹

Question	Tip to answer
A=1 → R=0.[2]	
3(a) A security system has inputs: **A** = correct password, **B** = correct PIN, **C** = card blocked. Output **P** is True when the user enters the correct password **OR** correct PIN, **AND** the card is **NOT** blocked. Write the Boolean expression for P. **(3)**	Brackets + NOT C.
P = (A OR B) AND NOT C.[123]	
3(b) Explain why the brackets are needed in **P = (A OR B) AND NOT C**. **(3)**	Order matters.
Brackets force A OR B to be worked out first.[1]Without brackets, the system might combine the wrong parts first and change the meaning.[2]This could allow/deny access incorrectly.[3]	
4(a) A logic circuit has inputs A and B going into an AND gate. The AND output then goes into a NOT gate to produce output P. Write the expression for P. **(4)**	AND then NOT.
4(b) Complete the truth table for **P = NOT(A AND B)**. **(4)**	AND first, then flip.
00: A AND B =0 → P=1.[1] 01: 0 → 1.[2] 10: 0 → 1.[3] 11: 1 → 0.[4]	
6(a) A cinema app lets a user in if: they are **12 or older** AND they have **paid**. Variables: age (integer), paid(Boolean). Write an algorithm (OCR Exam Reference Language) that outputs **"ALLOW"** or **"DENY"**. **(6)**	Two checks joined by AND.
age = int(input("Enter age")) [1]paid = input("Paid? Y/N")[2] if paid == "Y" AND age >= 12 then[3] print("ALLOW")[4] else[5] print("DENY")[6] endif	
6(b) A game unlocks a reward if the player has **either** collected a key (**K**) **or** completed the quest (**Q**), and they are **not banned** (**B**). Write the Boolean expression, then complete the truth table for all 3 inputs. **(6)**	Use (K OR Q) AND NOT B.
Expression: P = (K OR Q) AND NOT B.[1] Truth table rows (KQB → P): 000 → 0.[2] 001 → 0.[3] 010 → 1.[4] 011 → 0.[5] 100 → 1.[6] 101 → 0.[6] 110 → 1.[6] 111 → 0.[6]	

02 KEY VOCABULARY / 2.4.1

Boolean logic and truth tables

Use the correct technical wording before attempting longer answers.

AND

Output 1 only when both inputs are 1.

OR

Output 1 when one or both inputs are 1.

NOT

Reverses a Boolean value.

Truth table

Lists every input combination and the output.

Boolean expression

Combines logical variables and operators into a condition.

03 WORKED EXAMPLE / 2.4.1

Boolean logic and truth tables

Follow a complete example. Notice the method before attempting the separate questions.

Task
Evaluate P = (A OR B) AND NOT A.

Think it through — small steps

1. List every two-input combination: 00, 01, 10, 11.
2. For 00, A OR B is 0 so P=0.
3. For 01, A OR B is 1 and NOT A is 1 so P=1.
4. For 10 and 11, NOT A is 0 so P=0.

Code / trace example

```
A B | A OR B | NOT A | P
0 0 |   0    |   1   | 0
0 1 |   1    |   1   | 1
1 0 |   1    |   0   | 0
1 1 |   1    |   0   | 0
```

Why this earns marks
Working columns show exactly how the final Boolean output was obtained.

SPEC 2.4.1 | SKILL 25 OF 28

AND, OR and NOT gates

REMEMBER

AND is true only when both inputs are true; OR is true when at least one is true; NOT reverses a Boolean value.

THINK OF IT LIKE...

AND is two keys; OR is either key; NOT flips a switch.

WORKED EXAMPLE / APPLY

TRUE AND FALSE = FALSE; TRUE OR FALSE = TRUE.

YOU DO - EXAM PRACTICE

Calculate the outputs of FALSE OR TRUE and NOT TRUE. (2)

Write your answer before opening the separate answer section.

SPEC 2.4.1 | SKILL 26 OF 28

Truth tables

REMEMBER

A truth table lists all combinations of Boolean inputs and the output of an expression.

THINK OF IT LIKE...

A checklist of every possible switch setting.

WORKED EXAMPLE / APPLY

For A AND NOT B, if A=1 and B=0, output 1.

YOU DO - EXAM PRACTICE

Evaluate A AND NOT B for A=1, B=0 and for A=1, B=1. (2)

Write your answer before opening the separate answer section.

04 INDEPENDENT EXAM PRACTICE / 2.4.1

Boolean logic and truth tables

Do this without viewing the next answer pages. Use a whiteboard first for code/traces.

EXAM-STYLE CHALLENGE · 6 MARKS

For Q = (A AND NOT B) OR C, find Q for all four input rows shown. Explain the order of the NOT, AND and OR operations.

```
A B C | Q
0 0 0 | ?
1 0 0 | ?
1 1 0 | ?
0 1 1 | ?
```

Your response

PREDICT → WRITE → TEST → IMPROVE | Self-mark after finishing the entire topic.

MARK SCHEME | SKILL 25 OF 28

AND, OR and NOT gates

QUESTION

Calculate the outputs of FALSE OR TRUE and NOT TRUE. (2)

MODEL ANSWER

FALSE OR TRUE = TRUE (1). NOT TRUE = FALSE (1).

SELF-MARK

Give credit for equivalent correct wording. Correct your answer in a different colour, then revisit the study card if a key point was missing.

MARK SCHEME | SKILL 26 OF 28

Truth tables

QUESTION

Evaluate A AND NOT B for A=1, B=0 and for A=1, B=1. (2)

MODEL ANSWER

When A=1, B=0: NOT B=1, output 1 (1). When A=1, B=1: NOT B=0, output 0 (1).

SELF-MARK

Give credit for equivalent correct wording. Correct your answer in a different colour, then revisit the study card if a key point was missing.

05 ANSWERS & MARK SCHEME / 2.4.1

Boolean logic and truth tables

This answer belongs to the exam-style challenge on the previous practice page.

6-mark challenge · Model answer
Evaluate a three-input expression

Mark 1 For A=0 B=0 C=0, Q=0.

Mark 2 For A=1 B=0 C=0, Q=1.

Mark 3 For A=1 B=1 C=0, Q=0.

Mark 4 For A=0 B=1 C=1, Q=1.

Mark 5 NOT B is calculated before AND and OR.

Mark 6 The AND is evaluated before the final OR, following parentheses.

Accept equivalent correct methods, code or relevant wording. Award each point once.

WORKED SOLUTION / 2.4.1

Boolean logic and truth tables

See a complete method and compare it with each mark-scheme point.

Expression: Q = (A AND NOT B) OR C
Calculate NOT first, then AND, then the final OR.

A	B	C	NOT B	A AND NOT B	Q
0	0	0	1	0	0
1	0	0	1	1	1
1	1	0	0	0	0
0	1	1	0	0	1

Marking reminder
For multi-stage logic, working columns make the Boolean operations transparent and prevent accidental use of the wrong precedence.